

Prognostic and Health Management of Workstations Using Machine Learning

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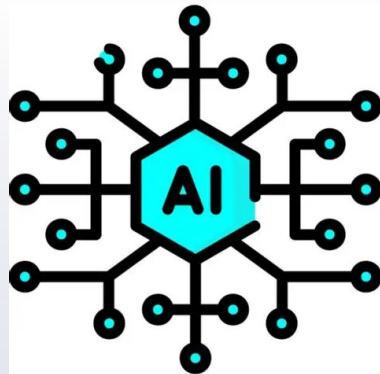
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General Context

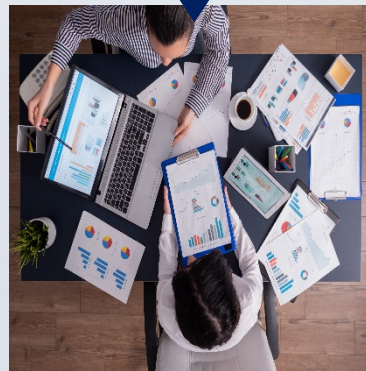
Limited visibility on
workstation health
before a failure
occurs
Manual diagnostic



Unexpected failures
and SLA risks



Largely available
data (logs, metrics)
underutilized for
failure management

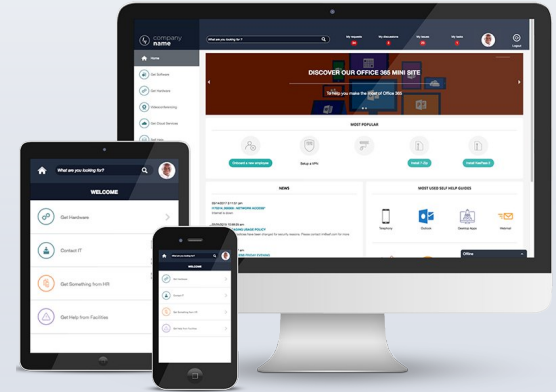


Reducible electronic
waste

Scientific Context and Research Challenges



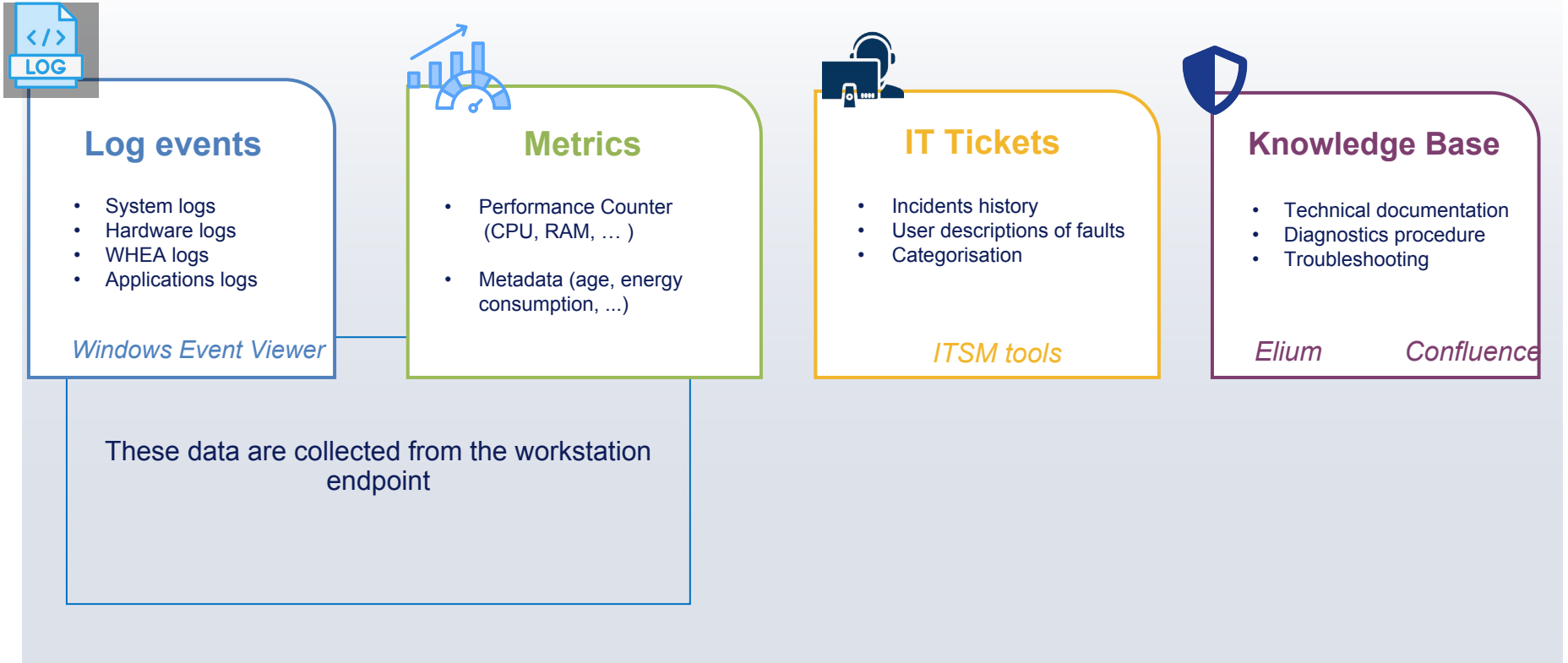
- w **Limited observability**: absence of dedicated mechanisms for collecting workstation logs and performance metrics
- w **Data quality limitations**: missing labels and incomplete records
- w **Integration and alignment** of heterogeneous data sources (logs, metrics, and incident tickets)
- w Challenges in conducting **Root Cause Analysis** (RCA) due to insufficient data quality



Research Questions

- How can **heterogeneous** data be correlated to detect **weak early-warning** signals of **workstation failures** ?
- How can incident tickets be leveraged to identify root causes of workstation failures ?

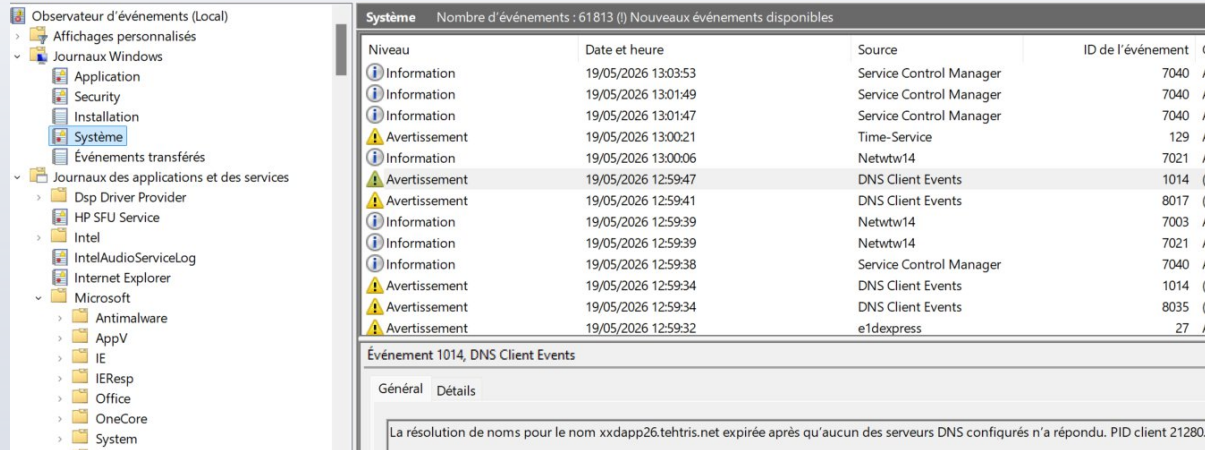
What heterogeneous data ?



Testbed Fault Injection (Ongoing Work)

— Context

- Injecting controlled and reproducible failures in order to identify weak warning signals



The screenshot displays the Windows Event Viewer interface. On the left, the 'System' log is selected under 'Journaux Windows'. The main pane shows a list of system events. The table below represents the data visible in the 'Details' pane for event ID 1014.

Niveau	Date et heure	Source	ID de l'événement	Catégorie
Information	19/05/2026 13:03:53	Service Control Manager	7040	A
Information	19/05/2026 13:01:49	Service Control Manager	7040	A
Information	19/05/2026 13:01:47	Service Control Manager	7040	A
Avertissement	19/05/2026 13:00:21	Time-Service	129	A
Information	19/05/2026 13:00:06	Netwtw14	7021	A
Avertissement	19/05/2026 12:59:47	DNS Client Events	1014	(1)
Avertissement	19/05/2026 12:59:41	DNS Client Events	8017	(1)
Information	19/05/2026 12:59:39	Netwtw14	7003	A
Information	19/05/2026 12:59:39	Netwtw14	7021	A
Information	19/05/2026 12:59:38	Service Control Manager	7040	A
Avertissement	19/05/2026 12:59:34	DNS Client Events	1014	(1)
Avertissement	19/05/2026 12:59:34	DNS Client Events	8035	(1)
Avertissement	19/05/2026 12:59:32	e1dexpress	27	A

Below the table, the details for 'Événement 1014, DNS Client Events' are shown. The 'Général' tab is active, displaying the message: 'La résolution de noms pour le nom xxdapp26.tehtris.net expirée après qu'aucun des serveurs DNS configurés n'a répondu. PID client 21280.'

Log events from Windows Event Viewer

Testbed Experiments (Ongoing Work)

- Fault Injections (CPU, Memory, Disk, network packet loss, network latency)

Chaos Experiments			
Project Settings >			
Account Settings ⓘ			
Organization Settings ⓘ			
EXPERIMENT NAME	DETAILS	RECENT EXPERIMENT RUNS	LAST MODIFIED
☐ disk-stress 1 ID: cbe0ea7-1741-4a0...	▶ Non-Cron dell-physique		amadou kabir 6 Aug 2026, 15:02
☐ network-loop 1 ID: 3a0d60fa-8921-4f...	▶ Non-Cron dell-physique		amadou kabir 28 Jul 2026, 16:07
☐ memory-stress 1 ID: 54bdf72-1b83-43...	▶ Non-Cron dell-physique		amadou kabir 27 Jul 2026, 18:02
☐ network 1 ID: 791670c2-2c36-4c...	▶ Non-Cron dell-physique		amadou kabir 28 Jul 2026, 14:16

Harness Chaos Engineering

Natural Language Processing (NLP)



Goal

- Leveraging IT incident tickets for automated diagnosis
- Predicting the root cause from user ticket descriptions
- Evaluation is based on performance metrics: precision, recall, f1-score, accuracy

Preliminary Results

- Initial tests on a real IT ticket dataset show that user ticket descriptions alone provide limited information for incident classification

Modèle	Accuracy	Precision	Recall	F1-Macro	F1-Weighted
SVM (Linear)	0.504	0.408	0.371	0.381	0.492
Logistic Regression	0.532	0.579	0.335	0.342	0.501
DistilCamemBERT	0.441	0.344	0.458	0.364	0.412



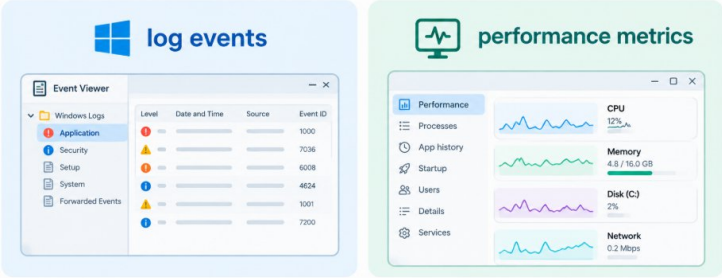
Evaluated Methods

- **Data Exploration:** Topic modeling, cosine similarity
- **Tokenization et Embeddings:** BoW, TF-IDF, Word2Vec, BertTokenizer
- **Classifiers:** SVM, XGBoost, LR, CamemBERT, FlauBERT, DistilCamemBERT



Future Work

- Testing the hypothesis that semantic enrichment may improve incident ticket classification by adding technical additional context from logs and metrics



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